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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Design and Analysis of Algorithms (course)

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Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 :
Introduction
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Week 1 :
Analysis of
algorithms ()

Week 1 Quiz
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Week 2 :
Searching
and sorting
()

Week 2 Quiz

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

As per our records you have not submitted this assignment.

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) Consider the following variation of the merge function, where the input lists A and B are assumed to be sorted, with no duplicated elements, and C is the output list. **2 points**

```
function StrangeMerge(A,m,B,n,C) {
    // A has m elements, B has n elements
    i = 0; j = 0; k = 0;

    while (i+j < m+n) {
        if (i == m) {C[k] = B[j]; j++; k++;}
        if (j == n) {C[k] = A[i]; i++; k++;}

        if (i < m and j < n) {
            if (A[i] < B[j]) {C[k] = A[i]; i++; k++;}
            if (A[i] == B[j]) {i++; j++;}
            if (A[i] > B[j]) {C[k] = B[j]; j++; k++;}
        }
    }
}
```

What does C contain after executing StrangeMerge(A,m,B,n,C)?

- ☐ C contains the intersection of A and B.
- ☐ C contains all values in A that do not occur in B.
- ☐ C contains all values in B that do not occur in A.
- ☐ C contains all values that occur in either A or B, but not in both input lists.

Week 2 Quiz ()

Quiz: Week 2 Quiz
(assessment? name=235)

Week 2 Programming Assignment ()

Week 3 : Graphs ()

Week 3 Quiz ()

Week 3 Programming Assignment ()

Week 4 : Weighted graphs ()

Week 4 Quiz ()

Week 4 Programming Assignment ()

Week 5: Data Structures: Union-Find and Heaps ()

Week 5 : Divide and Conquer ()

Week 5 Quiz ()

No, the answer is incorrect.

Score: 0

Feedback:

The merge skips any element that occurs in both A and B. So list C contains all elements that occur in A but not in B or vice versa, sometimes called the symmetric difference of A and B.

Accepted Answers:

C contains all values that occur in either A or B, but not in both input lists.

2) Suppose we modify mergesort as follows. We split the input into three equal segments, recursively sort each segment and then do a three way merge of the three sorted segments to obtain a fully sorted output. **2 points**

If we work out the recurrence for this variant, we find that the worst case complexity is $O(n \log_3 n)$, as opposed to $O(n \log_2 n)$ for the usual divide and conquer that splits the input into two equal segments.

Let us call the new version 3-way mergesort. What can we say about the asymptotic worst case complexity of 3-way-mergesort with the usual mergesort?

- ☐ The asymptotic worst case complexity of 3-way mergesort is the same as that of usual mergesort.
- ☐ The asymptotic worst case complexity of 3-way mergesort is better than that of usual mergesort.
- ☐ The asymptotic worst case complexity of 3-way mergesort is worse than that of usual mergesort.
- ☐ This depends on the length and the initial arrangement of values of the input.

No, the answer is incorrect.

Score: 0

Feedback:

A direct recursive analysis yields a complexity $O(n \log_3 n)$. However, $\log_3 n$ is $\log_2 3 \log_2 n$, so the asymptotic complexity is the same as regular merge sort.

Accepted Answers:

The asymptotic worst case complexity of 3-way mergesort is the same as that of usual mergesort.

3) Suppose a new generation CPU can process 10^{10} operations per second. You have to sort an array with 10^8 elements. Which of the following is true? **2 points**

- ☐ Insertion sort will always take several hours while merge sort will always take less than 1 second.
- ☐ Quicksort will always take several hours while merge sort will always take less than 1 second.
- ☐ Insertion sort could take several hours while merge sort will always take less than 1 second.
- ☐ Insertion sort could take several hours while quicksort will always take less than 1 second.

Week 6: Data Structures: Search Trees ()

Week 6: Greedy Algorithms ()

Week 6 Quiz ()

Week 6 Programming Assignment ()

Week 7: Dynamic Programming ()

Week 7 Quiz ()

Week 7 Programming Assignment ()

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No, the answer is incorrect.

Score: 0

Feedback:

Mergesort has worst case complexity $O(n \log n)$, so all inputs will take less than 1 second. Both insertion sort and quicksort have worst case complexity $O(n^2)$. However, this does not mean that all inputs will take time proportional to n^2 .

Accepted Answers:

Insertion sort could take several hours while merge sort will always take less than 1 second

4) Which of the following statements is **not** true about quicksort?

2 points

- ☐ For every fixed strategy to choose a pivot for quicksort, we can construct a worst case input that requires time $O(n^2)$.
- ☐ If we could find the median in time $O(n)$, quicksort would have worst case complexity $O(n \log n)$.
- ☐ Quicksort and merge sort are both examples of divide and conquer algorithms.
- ☐ If we randomly choose a pivot element each time, quicksort will always terminate in time $O(n \log n)$.

No, the answer is incorrect.

Score: 0

Feedback:

Choosing a random pivot only provides an expected (not worst case) running time of $O(n \log n)$.

Accepted Answers:

If we randomly choose a pivot element each time, quicksort will always terminate in time $O(n \log n)$.

5) We have a list of three dimensional points [(7,8,1),(3,7,5),(6,4,1),(6,9,5),(0,5,2),(9,9,0)]. We sort these in ascending order by the third coordinate. Which of the following corresponds to a stable sort of this input? **2 points**

- ☐ [(9,9,0),(7,8,1),(6,4,1),(0,5,2),(3,7,5),(6,9,5)]
- ☐ [(9,9,0),(7,8,1),(6,4,1),(0,5,2),(6,9,5),(3,7,5)]
- ☐ [(0,5,2),(3,7,5),(6,4,1),(6,9,5),(7,8,1),(9,9,0)]
- ☐ [(9,9,0),(6,4,1),(7,8,1),(0,5,2),(3,7,5),(6,9,5)]

No, the answer is incorrect.

Score: 0

Feedback:

Pairs with same third coordinate are (7,8,1) / (6,4,1) and (3,7,5) / (6,9,5). These must appear in the same order in the list sorted by third coordinate.

Accepted Answers:

[(9,9,0),(7,8,1),(6,4,1),(0,5,2),(3,7,5),(6,9,5)]